Section A – Answer any 2 out of 3 questions in section A

Question 1

a)

(i) If these three generators are used to supply 57 households, what is the probability to meet peak demand?

[6]

The availability table of generators:

Availability (kW)	*Probability (%)*
150	*91.26*
100	*8.47*
50	*0.2619*
0	*0.0027*

[4 marks]

The peak demand for 50 households is

$$\left(0.444 + \frac{1-0.444}{\sqrt{57}}\right) \times 57 \times 5 = 147.52\ kw$$

[1 mark]

The probability to meet peak demand is

91.26%

[1 mark]

(ii) If the reliability criterion requires that the probability to meet peak demand is above 99%, how many households can be supplied by these generators?

[4]

To meet 99% reliability requirement, the maximum peak demand can be served is 100 kW

[2 Mark]

$$\left(0.444 + \frac{1-0.444}{\sqrt{n}}\right) n \times 5 = 100$$

$$\therefore n = 37.4$$

37 households

[2 marks]

Most of the students answered this question correctly. A few had wrong answers solving the quadratic equation.

b)

(i) If there is a bid of 150MW of EFR, what is the maximum price (£/MW) the system operator should pay for it to achieve the minimum cost?

[6]

Model Answers

1) *Without EFR*

R=3375MW, EFR =0,

*cost 1= 3375*5=16875*

[2 mark]

2) *With EFR*

R=2858.268MW, EFR=150MW,

*cost 2= 2858.268*5+150*x=14291.339+150*x*

[2 mark]

To achieve minimum cost, cost1>=cost2, so the maximum price the operator would pay is 17.22 £/MW.

[2 marks]

Most students were able to find the cost of PFR when EFR is zero. However, some made mistakes finding the PFR and the associated cost when EFR is 150WM.

(ii) Discuss how the maximum price (£/MW) the system operator should pay for EFR would change if (1) the system inertia level (H) decreases and (2) the amount of EFR bid increases?

[4]

(1). The operator will pay higher maximum price for EFR if the system inertia level (H) decreases

[2 marks]

(2). The operator will pay lower maximum price for EFR if the amount of EFR bid increases

[2 marks]

Many students stated the right change directions. However, few were able to clearly demonstrate the reasons, which should be based on evaluating the partial derivatives.

Model Answers

Question 2

a)

To avoid the confusion caused by:
Multiple voltage levels
Impedance of transformers depends on side
Single-phase vs three phase
Some students did not give the reasons of all three aspects.

[1 mark each, 3]

b)

(i) Calculate the active and reactive powers in per unit at the receiving end given $\boldsymbol{V_s = 1\angle 0°}$, $\boldsymbol{V_r = 1.1\angle -5°}$ and $\boldsymbol{R + jX = 0.01 + j0.2}$, $\boldsymbol{jB = j0.02}$.

$$S = V_r i_r^* = P_r + jQ_r = V_r \left(\frac{V_s - |V_r|\angle\theta}{R + jX} - |V_r|\angle\theta \frac{B}{2} \angle 90^\circ \right)^*$$

[1 mark]

$$= V_r \left(\frac{|V_s| - |V_r|cos\theta - j|V_r|sin\theta}{R + jX} - \frac{|V_r|B}{2} \angle(\theta + 90^\circ) \right)^*$$

$$= (|V_r|cos\theta + j|V_r|sin\theta) \left[\left(\frac{|V_s|R - |V_r|(Rcos\theta + Xsin\theta)}{R^2 + X^2} + \frac{|V_r|B}{2} sin\theta \right) + j \left(\frac{|V_s|X + |V_r|(R\,sin\,\theta - X\,cos\theta)}{R^2 + X^2} + \frac{|V_r|B}{2} cos\theta \right) \right]$$

$$= 0.4497 - j0.5813$$

[2 marks]

Many students made mistakes on calculating the current from the shunt capacitor.

(ii) Repeat the calculation if $\boldsymbol{V_r = 0.9\angle -5°}$, and comment on the influence of the difference in voltage magnitude on the power flow.

$$S = V_r i_r^* = P_r + jQ_r = V_r \left(\frac{V_s - |V_r|\angle\theta}{R + jX} - |V_r|\angle\theta \frac{B}{2} \angle 90^\circ \right)^*$$

$$= V_r \left(\frac{|V_s| - |V_r|cos\theta - j|V_r|sin\theta}{R + jX} - \frac{|V_r|B}{2} \angle(\theta + 90^\circ) \right)^*$$

$$= (|V_r|cos\theta + j|V_r|sin\theta) \left[\left(\frac{|V_s|R - |V_r|(Rcos\theta + Xsin\theta)}{R^2 + X^2} + \frac{|V_r|B}{2} sin\theta \right) + j \left(\frac{|V_s|X + |V_r|(R\,sin\,\theta - X\,cos\theta)}{R^2 + X^2} + \frac{|V_r|B}{2} cos\theta \right) \right]$$

$$= 0.4128 + j0.4203$$

Model Answers

[1 mark]

Some had wrong results on calculations or expressions.

R<<X, difference in voltage magnitude is mainly related to reactive power flow

[1 mark]

The relationship between $|V_s|$ and $|V_r|$ influences the direction of the reactive power flow.

[1 mark]

Most students know the impact of voltage magnitude on relative power. However, the relationship between voltage magnitude difference and the relative power flow direction was overlooked.

(iii) Repeat the calculation if $\boldsymbol{V_r = 1.1\angle +5°}$, and comment on the influence of the difference in voltage angle on the power flow.

$$S = V_r i_r^* = P_r + jQ_r = V_r\left(\frac{V_s - |V_r|\angle\theta}{R + jX} - |V_r|\angle\theta\frac{B}{2}\angle 90^\circ\right)^*$$

$$= V_r\left(\frac{|V_s| - |V_r|cos\theta - j|V_r|sin\theta}{R + jX} - \frac{|V_r|B}{2}\angle(\theta + 90^\circ)\right)^*$$

$$= (|V_r|cos\theta + j|V_r|sin\theta)\left[\left(\frac{|V_s|R - |V_r|(Rcos\theta + Xsin\theta)}{R^2 + X^2} + \frac{|V_r|B}{2}sin\theta\right) + j\left(\frac{|V_s|X + |V_r|(R\,sin\,\theta - X\,cos\theta)}{R^2 + X^2} + \frac{|V_r|B}{2}cos\theta\right)\right]$$

$$= -0.5066 - j0.5335$$

[1 mark]

R<<X, difference in voltage angle is mainly related to active power flow.

[1 mark]

The relationship between $\angle V_s$ and $\angle V_r$ influences the direction of the active power flow.

[1 mark]

Similarly, most students know the impact of voltage angle on active power. However, the relationship between voltage angle and the active power flow direction was overlooked.

c)

The expressions of currents and voltages for a three-phase system are as follows.

$v_a(t) = \sqrt{2}Vsin\omega t\,,\ i_a(t) = \sqrt{2}Isin(\omega t - \phi)$

$v_b(t) = \sqrt{2}Vsin(\omega t - 120^o)\,,\ i_b(t) = \sqrt{2}Isin(\omega t - \phi - 120^o)$

$v_c(t) = \sqrt{2}Vsin(\omega t - 240^o)\,,\ i_c(t) = \sqrt{2}Isin(\omega t - \phi - 240^o)$

[2 mark each, 6]

Model Answers

Some students made mistakes on phase difference and some forgot to include the frequency (ωt).

$$p_3(t) = 3VI\cos\phi = 3 * 1 * 0.5 * 0.5 = 0.75\ pu$$

[1 mark]

Some forgot to account for three phases.

$$q_3(t) = 0$$

[1 mark]

Model Answers

Question 3

a)

At a Slack Bus:

It serves as the voltage angle reference. As Plosses are unknown before calculation, slack bus is used to balance load and generation.

[1 mark]

At a PV bus

- *P is specified*
- *|V| is specified*
- *Voltage angle is unknown*
- *Q is unknown*
- *Q adjusted to keep |V| constant*
- *Most generator busses are PV busses*

[1 mark]

At a PQ bus

- *P is specified*
- *Q is specified*
- *Voltage magnitude and angle are unknown*
- *Most load busses are PQ busses*

[1 mark]

Most students answered the questions correct.

b)

$$i_1 = \frac{1}{Z_1}(V_1 - V_2) + \frac{1}{Z_2}(V_1 - V_3) = \left(\frac{1}{Z_1} + \frac{1}{Z_2}\right)V_1 - \frac{1}{Z_1}V_2 - \frac{1}{Z_2}V_3$$

$$i_2 = \frac{1}{Z_1}(V_2 - V_1) + \frac{1}{Z_3}(V_2 - V_3) = -\frac{1}{Z_1}V_1 + \left(\frac{1}{Z_1} + \frac{1}{Z_3}\right)V_2 - \frac{1}{Z_3}V_3$$

$$i_3 = \frac{1}{Z_2}(V_3 - V_1) + \frac{1}{Z_3}(V_3 - V_2) = -\frac{1}{Z_2}V_1 - \frac{1}{Z_3}V_2 + \left(\frac{1}{Z_2} + \frac{1}{Z_3}\right)V_3$$

[1 mark each, 3]

$$\therefore Y = \begin{pmatrix} \frac{1}{Z_1} + \frac{1}{Z_2} & -\frac{1}{Z_1} & -\frac{1}{Z_2} \\ -\frac{1}{Z_1} & \frac{1}{Z_1} + \frac{1}{Z_3} & -\frac{1}{Z_3} \\ -\frac{1}{Z_2} & -\frac{1}{Z_3} & \frac{1}{Z_2} + \frac{1}{Z_3} \end{pmatrix}$$

[2 marks]

Most students answered the questions correct.

c)

[12]

Model Answers

$V_1 = 1\angle 0°$ $V_2^0 = 1.05\angle 0°$ $V_3^0 = 1\angle 0°$

[1 mark]

$$Y = \begin{pmatrix} -j8 & j4 & j4 \\ j4 & -j9 & j5 \\ j4 & j5 & -j9 \end{pmatrix}$$

[1 mark]

Iteration 1:

$$i_3 = \frac{S_3^*}{V_3^*} = Y_{31}V_1 + Y_{32}V_2 + Y_{33}V_3$$

$$V_3^1 = \frac{1}{Y_{33}}\left(\frac{S_3^*}{V_3^{0*}} - Y_{31}V_1 - Y_{32}V_2^0\right) = \frac{1}{-j9}\left(\frac{-1+1.03j}{1} - j4 - j5 \times (1.05)\right)$$
$$= 0.9133 - j0.1111$$

[2 marks]

Some students made mistake on evaluating complex conjugate.

$$S_2^* = V_2^{0*} i_2 = V_2^{0*}(Y_{21}V_1 + Y_{22}V_2^0 + Y_{23}V_3^1)$$
$$= 1.05 \times (j4 - j9 \times 1.05 + j5 \times (0.9133 - j0.1111))$$
$$= 0.5833 - j0.9277$$

$$Q_2 = j0.9277$$

[2 marks]

Few students did not discard the active power.

$$V_2^1 = \frac{1}{Y_{22}}\left(\frac{S_2^*}{V_2^{0*}} - Y_{21}V_1 - Y_{23}V_3^1\right) = \frac{1}{-j9}\left(\frac{1 - j0.9277}{1.05} - j4 - j5(0.9133 - j0.1111)\right)$$
$$= 1.05\angle 2.4049°$$

[2 marks]

Iteration 2:

$$V_3^2 = \frac{1}{Y_{33}}\left(\frac{S_3^*}{V_3^{1*}} - Y_{31}V_1 - Y_{32}V_2^1\right)$$
$$= \frac{1}{-j9}\left(\frac{-1+1.03j}{0.9133 + j0.1111} - j4 - j5 \times (1.05\angle 2.4049°)\right)$$
$$= 0.8928 \angle -5.1657°$$

[1 mark]

Few did not use the value of V2 from previous iteration.

$$S_2^* = V_2^{1*} i_2 = V_2^{1*}(Y_{21}V_1 + Y_{22}V_2^1 + Y_{23}V_3^2)$$
$$= 1.05\angle$$
$$-2.4049° \times (j4 - j9 \times 1.05\angle 2.4049° + j5 \times (0.8928 \angle -5.1657°))$$
$$= 0.7938 - j1.0799$$

$$Q_2 = j1.0799$$

[1 mark]

Model Answers

$$V_2^2 = \frac{1}{Y_{22}}\left(\frac{S_2^*}{V_2^{1*}} - Y_{21}V_1 - Y_{23}V_3^2\right)$$
$$= \frac{1}{-j9}\left(\frac{1 - j1.0799}{1.05\angle -2.4049°} - j4 - j5(0.8928 \angle -5.1657°)\right)$$
$$= 1.05\angle 3.5956°$$

[2 marks]

Some students made mistake on evaluating complex conjugate.

Model Answers

Question 4

a)

At fault bus m, the passive network is excited with $-V_m^0$ in series with fault impedance Z_f
Change in bus voltages due to fault

$$\Delta V_{BUS} = Z_{BUS} \times I_{BUSf}$$
$$I_{BUSf} = \begin{bmatrix} 0 & 0 & \cdots & I_m(=-I_f) & \cdots & 0 \end{bmatrix}^T$$

[1 mark]

$$\Delta V_{BUS} = \begin{bmatrix} \Delta V_1^0 \\ \Delta V_2^0 \\ \vdots \\ \Delta V_m^0 \\ \vdots \\ \Delta V_n^0 \end{bmatrix} = \begin{bmatrix} Z_{11} & Z_{12} & \cdots & Z_{1m} & \cdots & Z_{1n} \\ Z_{21} & Z_{22} & \cdots & Z_{2m} & \cdots & Z_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ Z_{m1} & Z_{m2} & \cdots & Z_{mm} & \cdots & Z_{mn} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ Z_{n1} & Z_{n2} & \cdots & Z_{nm} & \cdots & Z_{nn} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \vdots \\ -I_f \\ \vdots \\ 0 \end{bmatrix} = \begin{bmatrix} -Z_{1m}I_f \\ -Z_{2m}I_f \\ \vdots \\ -Z_{mm}I_f \\ \vdots \\ -Z_{nm}I_f \end{bmatrix}$$

[2 marks]

$$\begin{bmatrix} V_1^f \\ V_2^f \\ \vdots \\ V_m^f \\ \vdots \\ V_n^f \end{bmatrix} = \begin{bmatrix} V_1^0 \\ V_2^0 \\ \vdots \\ V_m^0 \\ \vdots \\ V_n^0 \end{bmatrix} + \begin{bmatrix} \Delta V_1^0 \\ \Delta V_2^0 \\ \vdots \\ \Delta V_m^0 \\ \vdots \\ \Delta V_n^0 \end{bmatrix} = \begin{bmatrix} V_1^0 - Z_{1m}I_f \\ V_2^0 - Z_{2m}I_f \\ \vdots \\ V_m^0 - Z_{mm}I_f \\ \vdots \\ V_n^0 - Z_{nm}I_f \end{bmatrix}$$

[1 mark]

Many students didn't show the first two steps (above) and directly used the Thevenin equivalent.

$$V_m^f = V_m^0 - Z_{mm}I_f = Z_f I_f$$
$$I_f = \frac{V_m^0}{Z_{mm} + Z_f}$$

[1 mark]

Voltage at bus k during fault

$$V_k^f = V_k^0 - Z_{km}I_f$$
$$= V_k^0 - V_m^0\left(\frac{Z_{km}}{Z_{mm} + Z_f}\right)$$

[2 marks]

b)

Three phase fault current at bus 1

Model Answers

$$I_1^f = \frac{V_1^0}{Z_{11} + Z_f} = \frac{1}{0.14 + 0.02} = 6.25 \text{ p.u.}$$

[2 marks]

Voltage at bus 2 during three-phase fault at bus 1

$$V_2^f = V_2^0 - Z_{21} I_1^f = 1 - 0.11 \times 6.25 = 0.313 \text{ p.u.}$$

[2 marks]

Voltage at bus 3 during three-phase fault at bus 1

$$V_3^f = V_3^0 - Z_{31} I_1^f = 1 - 0.13 \times 6.25 = 0.188 \text{ p.u.}$$

[2 marks]

Short circuit level at bus 3

$$SCL_3 = \frac{S_{base}}{Z_{TH}} = \frac{S_{base}}{Z_{33}} = \frac{500}{0.18} = 2777.8 \text{ MVA}$$

[2 marks]

In the exam question, the fault impedance was specified as 0.2 pu without specifying whether it's resistive or reactive. Some students used it as a resistance while some took it as reactance. Full mark was awarded for both approaches.

c)

- Electric circuit transients – Instantaneous current could be up to two times the steady-state value immediately after the fault due to the decaying DC offset introduced by the RL transients. Fault calculation is based on steady-state and such transients are considered empirically in circuit breaker design for example.

[1.5 marks]

Some students strayed into a general discussion around transients without being explicit about the doubling effect and how it's accounted for in an empirical way.

- Magnetic circuit transients – Generators and motors present a lower reactance immediately after the fault as the excess flux due to the fault current can't flow through the rotor or damper windings immediately. This is accounted for by using the sub-transient reactance (lower than the steady-state reactance) for fault calculation.

[1.5 marks]

Most students touched upon the constant flux linkage effect, but few closed the argument on use of sub-transient reactance.

d)

- Wind and solar PV are usually interfaced to the power grid through power converters which have very limited over current capability.

Model Answers

- Due to limited fault current contribution from such converter-interfaced generation sources, the short circuit level of the system declines when synchronous generators are replaced by them.

[1 mark each – 2 marks]

Model Answers

Question 5

a)

Equivalent sequence impedance at fault point

$$Z_{1eq} = j\left(X_{G1} + X_{T1} + {}^{X_{L1}}/_{2}\right) \parallel j\left(X_{G1} + X_{T1} + {}^{X_{L1}}/_{2}\right) = j0.14 \text{ p.u.}$$

$$Z_{2eq} = j\left(X_{G2} + X_{T2} + {}^{X_{L2}}/_{2}\right) \parallel j\left(X_{G2} + X_{T2} + {}^{X_{L2}}/_{2}\right) = j0.14 \text{ p.u.}$$

A few students forgot to take the parallel equivalent.

The generator won't appear in the zero-sequence network due to delta connection of the transformer winding on the generator side

$$Z_{0eq} = j\left(X_{T0} + {}^{X_{L0}}/_{2}\right) \parallel j\left(X_{T0} + {}^{X_{L0}}/_{2}\right) = j0.1875 \text{ p.u.}$$

[1 mark each – 3 marks]

Some students did not consider the transformer reactance in the zero-sequence network.

Fault current

$$I_f = \frac{3}{Z_{1eq} + Z_{2eq} + Z_{0eq}} = -j6.417 \text{ p.u.}$$

[1 mark]

Some students left the fault current in sequence domain i.e. forgot to multiply by 3

Base current

$$I_b = \frac{S_b}{\sqrt{3}V_b} = \frac{100}{\sqrt{3} \times 132} = 0.437 \text{ kA}$$

[1 mark]

Fault current magnitude

$$|I_f| \times I_b = 2.81 \text{ kA}$$

[1 mark]

b)

$$V_c = V_{a0} + \alpha V_{a1} + \alpha^2 V_{a2}$$
$$= \left(-Z_{0eq} \times \frac{I_f}{3}\right) + \alpha\left(-jZ_{2eq} \times \frac{I_f}{3}\right) + \alpha^2\left(1 - jZ_{1eq} \times \frac{I_f}{3}\right)$$
$$= \alpha^2 - \frac{I_f}{3}\left(\alpha^2 \times Z_{1eq} + \alpha \times Z_{2eq} + Z_{0eq}\right)$$
$$= -0.6016 + j0.8660 = 1.055\angle 124.79° \text{ p.u.}$$

[1 mark for each step - 4 marks]

Model Answers

Most students used the correct expression for V_c but made mistakes in obtaining the three sequence voltages

$|V_B| = 1.055 \times 132 = 139.19$ kV

[1 mark]

c)

If both windings of the transformers are star with neutral grounded, the generator zero sequence reactance would appear in the zero-sequence network. The other two sequence networks remain the same.

$$Z_{0eq}{}' = j\left(X_{G0} + X_{T0} + {}^{X_{L0}}/_{2}\right) \parallel j\left(X_{G0} + X_{T0} + {}^{X_{L0}}/_{2}\right) = j0.2025 \text{ p.u.}$$

[1 mark]

Fault current

$$I_f{}' = \frac{3}{Z_{1eq} + Z_{2eq} + Z_{0eq}{}'} = -j6.218 \text{ p.u.}$$

[1 mark]

Fault current magnitude

$|I_f{}'| \times I_b = 2.72$ kA

[1 mark]

$$V_c = \alpha^2 - \frac{I_f}{3}\left(\alpha^2 \times Z_{1eq} + \alpha \times Z_{2eq} + Z_{0eq}{}'\right)$$
$$= -0.6202 + j0.8660 = 1.065\angle 125.61° \text{ p.u.}$$

[2 marks]

As before, most students used the correct expression for V_c but made mistakes in obtaining the three sequence voltages

$|V_B| = 1.065 \times 132 = 140.61$ kV

[1 mark]

d)

The negative and zero sequence networks would play a part in balanced fault current calculation

Three-phase fault current

Model Answers

$$I_{f3p} = \frac{1}{Z_{1eq}} = -7.143 \text{ p.u.}$$

[2 marks]

A few students used a multiplying factor of 3 for the fault current

Three-phase fault current magnitude
$|I_{f3p}| \times I_b = 3.124$ kA

[1 mark]

Model Answers

Question 6

a)

Reactance between the generator terminal and the large power system

$X_{ts} = X_T + X_L = 0.26$ p.u.

[0.5 mark]

Voltage phase angle at generator terminal bus

$$\theta = \sin^{-1}\left(\frac{P \times X_{ts}}{|V_t| \times V_s}\right) = 19.16°$$

[1 mark]

Generator internal voltage is given by

$$E_i = V_t\angle\theta + jX_G \times \left(\frac{V_t\angle\theta - V_s\angle 0}{jX_{ts}}\right)$$
$|E_i| = 1.0567$ p.u.

[1.5 marks]

Most students failed to do this correctly and the error propagated on to the subsequent parts of this problem.

b)

Total reactance between generator internal voltage and large power system

$X_{is} = X_G + X_{ts} = 0.33$ p.u.

$$P_{max} = \frac{|E_i| \times V_s}{X_{is}} = 3.202 \text{ p.u.}$$

[1 mark]

Initial power angle

$$\delta_0 = \sin^{-1}\left(\frac{P \times X_{is}}{|E_i| \times V_s}\right) = 23.95°$$

[1 mark]

Model Answers

Using the equal area criterion

$$\int_{\delta_0}^{\delta_{cr}} (P - 0)d\delta = \int_{\delta_{cr}}^{\pi-\delta_0} (P_{max} \sin\delta - P)d\delta$$

[1 mark]

$$P(\delta_{cr} - \delta_0) = P_{max}[\cos\delta_{cr} - \cos(\pi - \delta_0)] - P(\pi - \delta_0 - \delta_{cr})$$

$$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = 1.549 \text{ rad} = 88.73°$$

[1 mark]

Critical clearing time

$$t_{cr} = \sqrt{\frac{4H(\delta_{cr} - \delta_0)}{2\pi f \times P_m}} = 273.6 \text{ ms}$$

[1 mark]

c)

$$P_1 = 1.2 \times P = 1.56 \text{ p.u.}$$

New steady-state power angle

$$\delta_1 = \sin^{-1}\frac{P_1}{P_{max}} = 29.16°$$

[1 mark]

Using the equal area criterion

$$\int_{\delta_0}^{\delta_1} (P_1 - P_{max}\sin\delta)d\delta = \int_{\delta_1}^{\delta_{max}} (P_{max}\sin\delta - P_1)d\delta$$

Model Answers

[2 marks]

Most students applied EAC correctly but messed up the subsequent simplification.

$$\Rightarrow \cos\delta_0 - \cos\delta_{max} = \frac{P_1}{P_{max}}(\delta_{max} - \delta_0)$$

$$\Rightarrow 0.8733 - \cos\delta_{max} = 0.4872 \times (\delta_{max} - 0.5089)$$

[2 marks]

d)

i) Series compensation

$P_{max1} = \frac{|E_i| \times V_s}{X_G + X_T + X_L(1-k)} = 3.774$ p.u.

[1 mark]

Some students applied the compensation over the total reactance instead of just the line reactance. This mistake affected part (e) of this question.

ii) Ideal phase compensation

Steady-state stability limit is not affected by ideal phase angle compensation
$P_{max2} = 3.202$ p.u.

[1 mark]

iii) Ideal mid-point shunt reactive compensation

Steady-state stability limit would be constrained by the generator and transformer reactance

$P_{max3} = \frac{|E_i| \times V_m}{X_G + X_T + {}^{X_L}/_2} = 5.155$ p.u.

[1 mark]

Model Answers

Some students halved the total reactance rather than halving just the line reactance.

e)

With a TCSC having series compensation level of 20%

$$P_{max1} = \frac{|E_i| \times V_s}{X_G + X_T + X_L(1-k)} = 3.774 \text{ p.u.}$$

Initial power angle

$$\delta_0 = \sin^{-1}\left(\frac{P}{P_{max1}}\right) = 20.15°$$

[1 mark]

Using the equal area criterion

$$\int_{\delta_0}^{\delta_{cr}} (P - 0)d\delta = \int_{\delta_{cr}}^{\pi-\delta_0} (P_{max}\sin\delta - P)d\delta$$

[1 mark]

$$P(\delta_{cr} - \delta_0) = P_{max}[\cos\delta_{cr} - \cos(\pi - \delta_0)] - P(\pi - \delta_0 - \delta_{cr})$$

$$\delta_{cr} = \cos^{-1}[(\pi - 2\delta_0)\sin\delta_0 - \cos\delta_0] = 1.549 \text{ rad} = 95.66°$$

[1 mark]

Critical clearing time

$$t_{cr} = \sqrt{\frac{4H(\delta_{cr} - \delta_0)}{2\pi f \times P_m}} = 295.4 \text{ ms}$$

[1 mark]

Model Answers